**Registration No.:** | | | | | | | | | | |

**Total Number of Pages: 02** **Course: B.Tech/IDD**
**Sub_Code: 23ES1002**

**1st Semester Regular/Back Examination: 2024-25**
**SUBJECT: BASIC ELECTRONICS**
**BRANCH(S): CE, CHEM, BIOTECH, BIOMED, AE, CSIT, AERO, AEIE, AME, AUTO, CSE, CIVIL, CSE, CSEAI, CSEAIML, CSEDS, CST, ECE, ETC, EE, EEE, ELECTRICAL & C.E, ELECTRICAL, IT, ME, MECH, METTA, MINING, MINERAL, MME, EEVDT, CSEIOT**
**Time: 3 Hours**
**Max Marks: 100**
**Q.Code: R583**
**Answer Question No.1 (Part-I) which is compulsory, any eight from Part-II and any two from Part-III.**
**The figures in the right hand margin indicate marks.**

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**Part-I**

**Q1** **Answer the following questions:** **(2 x 10)**

**a)** What is a p-n junction diode? How does a barrier field appear across a p-n junction?
**b)** Explain the necessity of a parallel resistor in clamper circuit.
**c)** Compare between a FET and a BJT.
**d)** Why the field - effect transistor is called a unipolar transistor?
**e)** Mention advantages and limitations of integrated circuits.
**f)** State the characteristic an ideal Op-Amp.
**g)** Discuss the concept of “Virtual ground” in Op-Amp.
**h)** Determine the binary equivalent of 26.25.
**i)** State and prove De Morgan’s theorem.
**j)** Perform the binary addition
100101 + 101 + 1101 + 100

**Part-II**

**Q2** **Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve)** **(6 x 8)**

**a)** The reverse saturation current of a Si p-n junction is 5 nA at 27°C. What are the static and dynamic resistances of the junction for an applied forward bias of 0.6 V?
**b)** Discuss the concept of Zener and avalanche breakdown.
**c)** Draw the common base circuit of a junction transistor. Sketch its output characteristics. Explain the active, saturation, and cutoff regions.
**d)** Explain the term transistor biasing. What are the factors determining the choice of the Q-point? Draw the circuit diagrams of fixed-bias and voltage divider bias arrangement of an n-p-n transistor in CE configuration. Mention the limitations and advantages of both arrangements.
**e)** When the channel of a JFET is said to be pinched off? Define the pinch-off voltage. What is the pinch off current? By applying Shockley’s current expression, draw the transfer characteristic curve of an n-channel JFET.

**f)** Draw the circuit diagram of a common emitter n-p-n transistor with the following parameters: $V_{BB}$ = 5V, $R_B$ = 100 KΩ, $R_C$ = 1 KΩ, $V_{CC}$ = 10V, $V_{BE}$ = 0.7 V, $I_{CO}$ = 0, β = 100. Find $I_B$ and $I_C$. Is the transistor operating in the saturation region? Justify.

**g)** Explain the principle of operation of a digital inverter.

**h)** Draw a non-inverting configuration using Op-Amp. Calculate the output voltage of the circuit for $\upsilon_i$ = 150 mV rms, $R_f$ = 75 KΩ and $R_i$ = 36 KΩ.

**i)** Calculate the output voltage for the circuit shown in figure-1 with inputs of $V_1$ = 40 mV rms and $V_2$ = 20 mV rms.

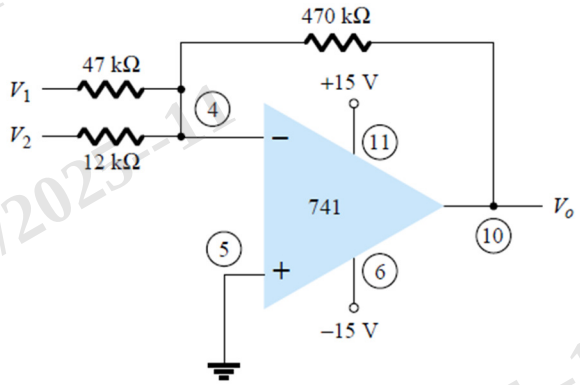



Figure-1

**j)** What is the difference between signed and unsigned numbers? Discuss the three different methods used for representing signed numbers in binary. Support each method by a suitable example.

**k)** Simplify the Boolean expressions to a minimum number of variables

I. $xy + x(wz + w\bar{z})$

II. $(x + \bar{y} + \bar{z})(\bar{x} + \bar{z})$

III. $\bar{A}B + \bar{A}\bar{C} + BC + A\bar{B}C$

**l)** I. Show that an EX-OR gate can be built with NOR gates.

II. How can the NAND gates be combined to perform the NOR operation?

## Part-III

**Only Long Answer Type Questions (Answer Any Two out of Four)**

**Q3** Draw the circuit diagram of (i) a half-wave rectifier and (ii) a full-wave rectifier. Explain the principle of operation of each circuit by applying a sinusoidal wave at the input of the rectifier. **(16)**

**Q4** What are different types of MOSFETs? Explain with a neat sketch, the structure and the working of an n-channel enhancement type MOSFET. Draw its drain and transfer characteristic curve. **(16)**

**Q5** Describe the use of operational amplifier as an integrator and differentiator. **(16)**

**Q6** $f_1 = AB + A\bar{C} + C + A\bar{B}C + ABC$ **(10 + 3 + 3)**

$f_2 = \bar{B} + ABC$

(i) Simplify the above functions. Draw the original and simplified circuit.

(ii) Express $f_1$ *and* $f_2$ in standard SOP form.

(iii) Express $f_1$ *and* $f_2$ in standard POS form.